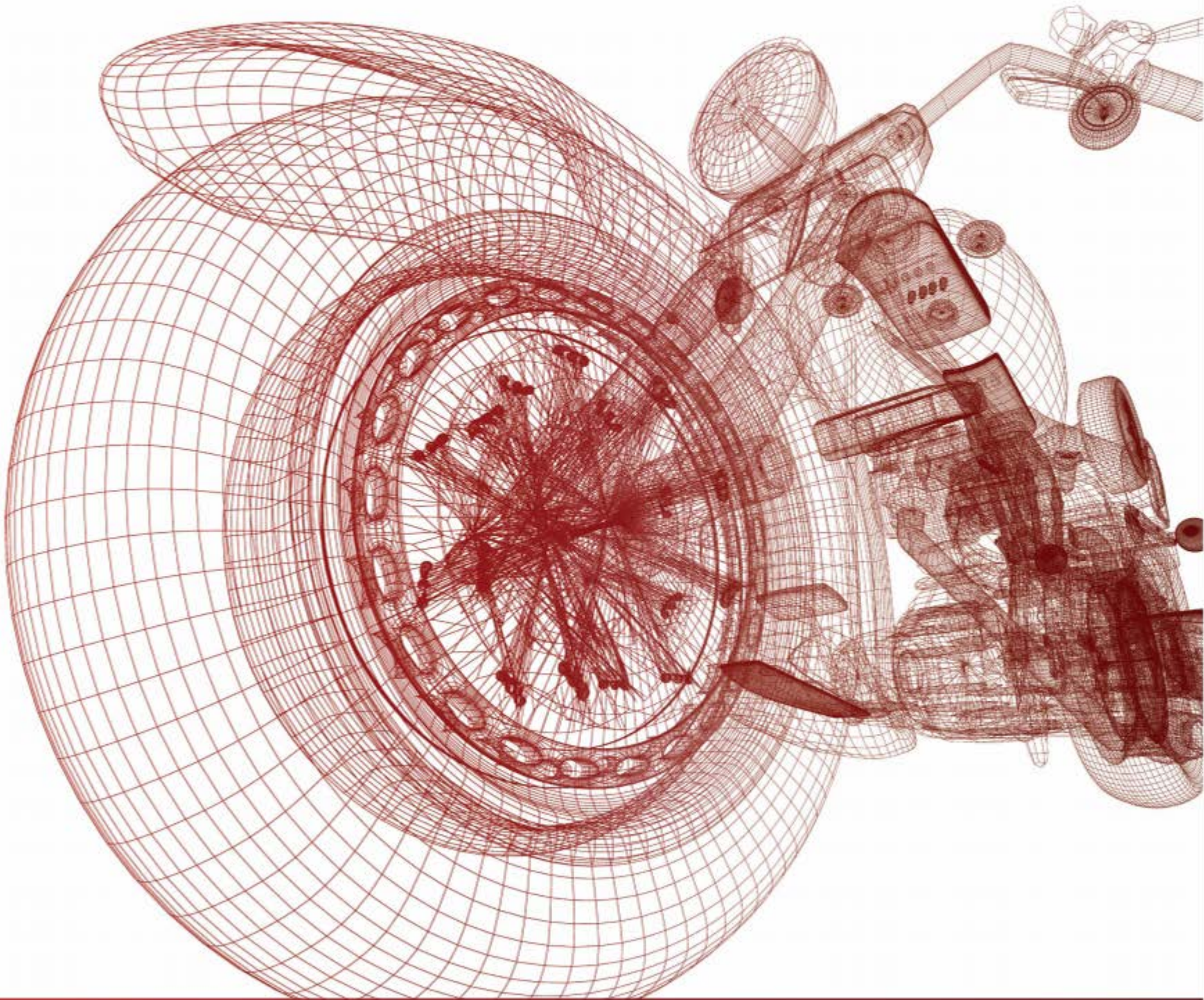




ASCENT®
CENTER FOR TECHNICAL KNOWLEDGE

Inventor 2022 - Advanced Assembly Modeling Review Questions - Set A



Autodesk® Inventor® 2022
Advanced Assembly Modeling
Learning Guide (Mixed Units) - 1st Edition

 **AUTODESK**
Authorized Publisher

Chapter Review Questions

1. What is the purpose of using an assembly folder?
 - a. To store component files.
 - b. To promote components.
 - c. To help organize an assembly.
 - d. To demote components.
2. Assembly folders have an impact on the relationships/degrees of freedom of the components in the folder.
 - a. True
 - b. False
3. What is the purpose of the *Constraint Set* tab shown in Figure 1–31?

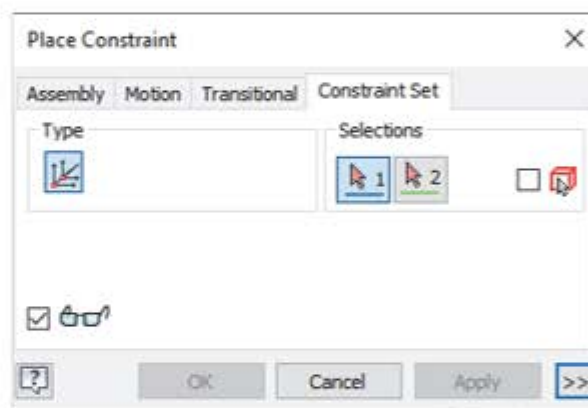


Figure 1–31

- a. Adds multiple constraints at the same time.
 - b. Enables you to constrain components using a user coordinate system (UCS).
 - c. Enables constraints to be grouped together.
 - d. None of the above.
4. $\begin{matrix} A \\ Z \end{matrix} \downarrow$ (Alpha Sort Component) sorts all of the components that exist at the top-level assembly and in subassemblies and assembly folders.
 - a. True
 - b. False

5. What do Motion constraints enable you to do?
 - a. Describe the movement of one surface relative to another.
 - b. Describe the constraints that keep components from moving.
 - c. Restrict the motion of components.
6. If a Rotation motion constraint enables one component to rotate when another rotates, what does the Rotation-Translation motion constraint do?
 - a. Enables two components to rotate when another rotates.
 - b. They are the same.
 - c. Enables one component to move translationally when another rotates.
 - d. Enables one component to move translationally when another moves translationally.
7. What do Transitional constraints enable you to do?
 - a. Describe the movement of one surface relative to a continuous set of surfaces.
 - b. Describe the movement of one surface relative to another.
 - c. Change constraint options depending on a component's other constraints.
 - d. Transition from one type of constraint to another.

Chapter Review Questions

1. What problems are associated with the bottom-up design approach? (Select all that apply.)
 - a. Interference or misalignment between components.
 - b. Incomplete design.
 - c. Modifications to components must be manually propagated throughout the assembly.
2. Which is not an advantage of the top-down design approach?
 - a. Organizes and helps enforce interactions and dependencies between components in an assembly.
 - b. Creates part geometry independent of the assembly or any other component.
 - c. Forces you to consider all areas of a final model before creating any geometry.
 - d. Helps create clean, reusable geometry that interacts as expected with the rest of the assembly.
3. Which is not a top-down design technique that is available as a specific tool in the Autodesk Inventor software?
 - a. Adaptive Modeling
 - b. Layout Design
 - c. Skeletal Modeling
 - d. Multi-Body Design
4. You can create 3D models from sketch blocks that were created in the Layout Design.
 - a. True
 - b. False

Chapter Review Questions

1. Additional feature geometry can be added to a model that has been created based on derived geometry from another model.
 - a. True
 - b. False
2. Which of the following items cannot be imported from the referenced part or assembly?
 - a. Parameters
 - b. Solid Bodies
 - c. Composite Features
 - d. Drawings
3. Match the symbols used to create a derived part (in the right column) with their definition shown in the left column.

Answer

- | | | |
|--|---|-------|
| a. The geometry is included in the derived part. |  | _____ |
| b. Some geometry is included while some is not included in the derived part. |  | _____ |
| c. The geometry is not included in the derived part. |  | _____ |

4. Match the symbols used to create a derived assembly (in the right column) with their definition shown in the left column.

Answer

a. The component is represented by a bounding box.



b. The volume is to be subtracted from the derived assembly.



c. The selected component intersects with the derived assembly.



d. The geometry is not included in the derived assembly.



e. The geometry is included in the derived assembly.



f. Some geometry is included while some is not included in the derived assembly.



5. When deriving a part model into a new part file, which of the following can you use to customize the derived geometry? (Select all that apply.)

- a. Select a specific Design View to derive from.
- b. Select parameters to include in the new file.
- c. Patch holes in the derived geometry.
- d. Specify a scale factor to set a size for the derived geometry.

6. A link that is broken with the base component can be unbroken; however, a suppressed link cannot be unsuppressed.

- a. True
- b. False

Chapter Review Questions

1. How do you activate the Representations browser?
 - a. From the **Model browser Representations** node.
 - b. From the Application Menu.
 - c. From the Model browser.
 - d. From the *View* tab in the ribbon.
2. You can place new components in the assembly when you have a user-defined Positional Representation active.
 - a. True
 - b. False
3. How do you override values to create a Positional Representation?
 - a. In an active Positional Representation, right-click on an object and select **Override**.
 - b. In an active Positional Representation, modify the required values.
 - c. Create a new assembly and modify the required values.
 - d. Switch to the Override setting in the ribbon.
4. Which of the following statements are true regarding Positional Representations? (Select all that apply.)
 - a. You can open an assembly so that a Positional Representation is automatically active in the model.
 - b. When a Positional Representation is active, you must explicitly save the assembly.
 - c. A copy of an existing Positional Representation enables you to start customizing its overrides based on the defaults in the Master.
 - d. Subassemblies within a top-level assembly can have Positional Representations that are specified for display in a top-level assembly.

5. While working with Positional Representations, which of the following can you perform using an Excel spreadsheet? (Select all that apply.)
 - a. Change values on previously overridden values.
 - b. Create new Positional Representations.
 - c. Delete a Positional Representation.
 - d. Rename a Positional Representation.
6. Once a constraint's offset value has been overridden in a Positional Representation, the Positional Representation must be deleted in order to remove the override.
 - a. True
 - b. False

Chapter Review Questions

1. Which is not a use of a simplified model in an assembly?
 - a. Incoming vendor models
 - b. Quickly suppressing all components in an assembly
 - c. Outgoing models
 - d. Large assembly management
2. Which of the following statements is true for creating a simplified model using the **Simplify** option?
 - a. If a view representation is active (with some component visibility turned off), any non-visible components are automatically set to be removed from the simplified model.
 - b. You must manually edit the surface geometry of the model to remove any unwanted holes after the simplified model is created.
 - c. The simplified model is non-associative with the parent geometry; it cannot be associative.
 - d. The iProperties for the simplified part update to reflect the actual geometry that was included in the simplified part.
3. Which of the following viewing options provide the ability to select components for inclusion again once they have been removed in a simplified view using the **Simplify View** option? (Select all that apply.)
 - a. **View All**
 - b. **View Included**
 - c. **View Excluded**
4. When an assembly is simplified using  (Simplify View), which of the following is created in the model and displays in the Model browser?
 - a. Envelope
 - b. Simple View
 - c. Model State
 - d. Positional Representation

5. The second row of options in the Envelope mini-toolbar, shown in Figure 9–36, enables you to do which of the following? (Select all that apply.)



Figure 9–36

- a. Set the bounding shape.
 - b. Set the selection priority.
 - c. Define the color of the envelope.
 - d. Determine if all instances of the selected component are also enveloped.
 - e. Define the selected component as hidden or visible in the simplified part.
6. Only a single component can be selected at one time to be replaced with an envelope using the  (Define Envelopes) command. The envelope can only represent the exact size of the selected component.
- a. True
 - b. False

Chapter Review Questions

1. The Model State functionality in an assembly model is mutually exclusive with which of the following Inventor assembly functionalities?
 - a. Design Views
 - b. iAssembly
 - c. Positional Representations
 - d. Derive Assemblies
2. Which of the following statements regarding assembly model states are true? (Select all that apply.)
 - a. Model states can be created in an assembly to vary part level dimensions.
 - b. Model states can be created in an assembly to vary assembly dimensions.
 - c. Model states are assigned the same iProperties as the Master model and cannot be changed.
 - d. The **Edit via Spreadsheet** option can be used to begin the creation of all model states and their configurations from the Master model state.
 - e. While in the context of a top-level assembly, part and subassembly model states can be activated, but not created.
 - f. To remove an assembly feature from an active model state, right-click and select **Delete** on the feature in the Model browser.
3. Once a drawing view has been created to represent a model state configuration, it must be deleted and recreated to switch configurations.
 - a. True
 - b. False
4. In your office environment, if you open an assembly file that was created for you and it has iAssemblies associated with it, will the Model States node be available?
 - a. Yes
 - b. No

5. If the Mounting Holes assembly feature is suppressed in the Vise-medium model state shown in Figure 10–48, will it be suppressed in the Vise-long model state?



Figure 10–48

- a. Yes
- b. No
6. A new substitute model state can be created in an assembly using which of the following substitution options? (Select all that apply.)
- a. **Derive Part**
 - b. **Derive Assembly**
 - c. **Simplify**
 - d. **Select Part File**
7. Which of the following commands can be used to quickly create a substitute part file for all subassemblies in a top-level assembly at once? (Select all that apply.)
- a. **Derive Part**
 - b. **Derive Assembly**
 - c. **Simplify**
 - d. **Create Derived Substitutes**

8. The **Link Model States** option automatically activates any subassembly model states that have the same name as the top-level assembly model state.
 - a. True
 - b. False
9. Which type of drawing annotation table is used to display the model state configuration settings in a drawing?
 - a. Parts List Table
 - b. Revision Table
 - c. General Table
 - d. Hole Table
10. Representations can be used in drawings but not in presentations.
 - a. True
 - b. False

Chapter Review Questions

1. What type of pattern can be created from a component in an assembly? (Select all that apply.)
 - a. Rectangular
 - b. Circular
 - c. Associative
2. When mirroring components in a subassembly, you can select the subassembly in the Model browser to automatically select all of its components.
 - a. True
 - b. False
3. Once components are patterned in an assembly, how can you make a single pattern element independent?
 - a. In the Model browser, double-click on the pattern element.
 - b. In the Model browser, right-click on the element and select **Suppress**.
 - c. In the Model browser, right-click on the pattern element and select **Independent**.
 - d. In the Model browser, select the pattern element and select **Independent** from the *Assembly* tab>Pattern panel.
4. Copied components are dependent on the source components.
 - a. True
 - b. False
5. Which tool can be used and still maintain constraints between components?
 - a. Mirror Components
 - b. Copy Components
 - c. Both Mirror Components and Copy Components

-
6. Which of the following statements is true regarding patterning components in an assembly? (Select all that apply.)
- a. The position of an independent pattern element is not controlled by pattern constraints.
 - b. All elements in a pattern can be set as independent.
 - c. Independent pattern elements can be reset as dependent; however, you must manually delete the independent copy that was generated.
 - d. To create an associative pattern, the component must be constrained to any one of the feature pattern instances.

Chapter Review Questions

1. To access the options in the panel shown in Figure 17–47, which of the feature groups must be enabled? (Select all that apply.)

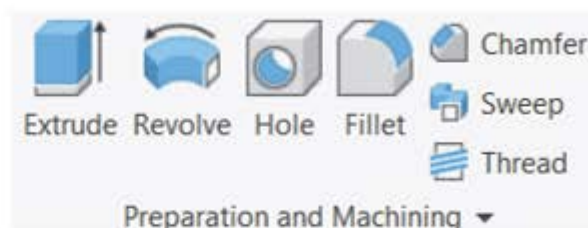


Figure 17–47

- a. Preparations
 - b. Welds
 - c. Machining
2. Which operations are performed before a weld is added?
- a. Preparations
 - b. Welds
 - c. Machining
3. Which type of weld is shown in Figure 17–48?

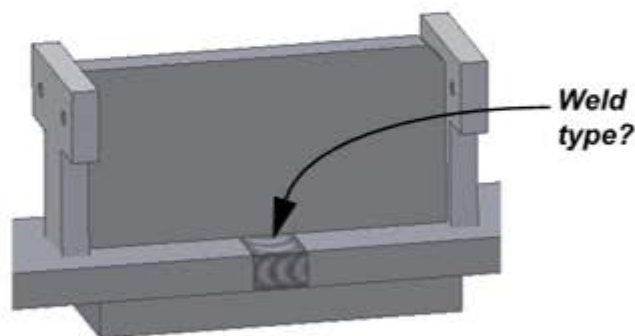


Figure 17–48

- a. Cosmetic Weld
 - b. Fillet Weld
 - c. Groove Weld
4. You can switch a normal assembly file to a weldment assembly, but you cannot switch a weldment file to a normal assembly file.
- a. True
 - b. False

5. Which of the following describe how you can create a new weld symbol? (Select all that apply)
 - a. In the *Weld* tab>Weld panel, click .
 - b. In the Fillet Weld dialog box, select **Create Welding Symbol** during weld creation.
 - c. Edit a Fillet Weld and select Create Welding Symbol.
 - d. Right-click on a weld in the Model browser and select **Create Welding Symbol**.
6. Which of the following are true statements about Cosmetic welds? (Select all that apply)
 - a. Cosmetic welds are created between two or more disconnected faces.
 - b. Cosmetic welds are located on one or more edges in the weldment assembly.
 - c. A Cosmetic weld must run the full length of the selected edge(s).
 - d. By default, Cosmetic welds are not included in the mass property calculations for the model.